

RE1 - Alternative resolution: closed loop with $S = A + B$

Data: $I_C = 14.26 \text{ mA}$, $g_m = 571 \text{ mS}$, $r_\pi = 350.5 \Omega$, $R_f = 50 \text{ k}\Omega$, $R_C = 300 \Omega$, $R_g = 1 \text{ k}\Omega$, $r_o = \infty$.

a) Topology and models of A and β

A senses the output voltage and subtracts input currents: **parallel-parallel** (current-voltage, transimpedance amplifier). Common variables v_1, v_2 , so both blocks are represented by the admittance matrix.

A =

$$\begin{bmatrix} 1/r_\pi & 0 \\ g_m & 1/R_C \end{bmatrix}$$

β (shunt R_f) =

$$\begin{bmatrix} 1/R_f & -1/R_f \\ -1/R_f & 1/R_f \end{bmatrix}$$

b) Gains A' and β'

Loading A with R_g and β ($S = A + B$, $S_{11g} = S_{11} + 1/R_g$):

$$A' = -S_{21} / (S_{11g} S_{22}) = -(g_m - 1/R_f) / [(1/r_\pi + 1/R_f + 1/R_g)(1/R_C + 1/R_f)] = \mathbf{-43.9 \text{ k}\Omega}$$

$$\beta' = S_{12} = -1/R_f = \mathbf{-20 \mu S}$$

$$\text{Consistency: } A'/(1+A'\beta') = -S_{21} / (S_{11g} S_{22} - S_{12} S_{21}) = -S_{21} / \det S_g$$

$$1 + A'\beta' = 1.879$$

c) V_o/V_s

Feedback theory: $A_f = A'/(1+A'\beta') = -23.39 \text{ k}\Omega$, $K_v = A_f/R_g = \mathbf{-23.39}$.

Same result with $S = A + B$, R_g and R_i excluded (ideal Norton source):

$$\begin{bmatrix} 1/r_\pi + 1/R_f & -1/R_f \\ g_m - 1/R_f & 1/R_C + 1/R_f \end{bmatrix}$$

$$\det S = S_{11} S_{22} - S_{12} S_{21} = 2.1044 \times 10^{-5} \text{ S}^2$$

$$v_2/i_1 = -S_{21}/\det S = -27.11 \text{ k}\Omega \text{ (magnitude } S_{21}/\det S, \text{ minus sign = inversion)}$$

$$K_v = (v_2/i_1)/(R_g + Z_i) = -27.11 \text{ k}\Omega / (1 \text{ k}\Omega + 159.3 \Omega) = \mathbf{-23.39}$$

Check with R_g included in S_{11} from the start: $K_v = -23.39$.

d) Z_i and Z_o

$$Z_i = S_{22}/\det S \text{ (} R_g \text{ excluded)} = \mathbf{159.3 \Omega}$$

$$Z_o = S_{11g}/\det S_g \text{ with } R_g \text{ included, } R_i = \infty = \mathbf{158.7 \Omega}$$

e) Frequency responses

Basic amplifier: $f_L = 500 \text{ Hz}$, $f_H = 400 \text{ kHz}$, mid-band $|A'_0| = 43.9 \text{ k}\Omega$. With feedback the mid-band gain drops to $23.39 \text{ k}\Omega$ and

$$f_{Lf} = f_L / (1+A'_0\beta') = \mathbf{266 \text{ Hz}}, f_{Hf} = f_H (1+A'_0\beta') = \mathbf{751 \text{ kHz}}.$$

Bandwidth: 399.5 kHz without feedback, 751.2 kHz with feedback.

Errors found in the original resolution (R1.pdf)

- $g_m = I_C/V_T = 14.26 \text{ mA} / 25 \text{ mV} = \mathbf{570.6 \text{ mS}}$, not 584 mS as printed (584 mS would need $I_C = 14.6 \text{ mA}$).

- $R_{\pi} = \beta_0 / g_m = 200 / 0.5706 = \mathbf{350.5 \Omega}$, not 542.5 Ω as printed. The later numbers in R1.pdf (e.g. $Z_{iA'} = 253.8 \Omega$) were obtained with 342.5 Ω ($= 200 / 0.584$), so they follow neither the printed r_{π} nor the correct one.

- Because of the wrong g_m / r_{π} , R1.pdf gives $A' = 44.7 \text{ k}\Omega$ (correct: 43.9 $\text{k}\Omega$), $A_f = K_v = -23.6$ (correct: -23.39), $Z_i = 154.8 \Omega$ (correct: 159.3 Ω), $Z_o = 157.5 \Omega$ (correct: 158.7 Ω).

- $f_{Hf} = 1364 \text{ kHz}$ in R1.pdf is wrong: $f_H(1 + A'_0 \beta') = 400 \text{ kHz} \times 1.879 = \mathbf{751 \text{ kHz}}$ (even with its own $1 + A'_0 \beta' = 1.894$ the result would be 757.6 kHz).